Duong Xuan Hieu

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EMPLOYMENT

AI Engineer **VNPT-IT** *05/2022 – Present*

- Developed Information Retrieval System for VNPT Smart-bot
  - Re-pretrained PhoBERT encoder to suitable for retrieval and semantic search tasks.
  - Optimized retrieval model by further fine-tuning and using knowledge distillation.
  - Developed and integrated ranking model to boost retrieval system performance.
  - Integrated the approximate nearest neighbor libraries such as Faiss for efficient search and easy to scale-up.

NLP Researcher **VNPT-IT** *10/2021 – 04/2022*

- Question Answering: Built a model to answer questions in the form of span extraction from knowledge document, the model has state-of-the-art performance on the VLSP-MRC 2021 dataset.
- Legal Text Retrieval: Built a retrieval model for legal text, my solution reached the top-3 public test and top-5 private test at ZaloAIChallenge 2021.

Data Scientist **MISA** *02/2021 – 04/2021*

- Built a model to classify customers based on purchase invoice information.

NLP Researcher **PTIT** *01/2021 – Present*

- Developed joint-model for natural language and spoken language understanding tasks such as: intent detection and slot filling.

EDUCATION

Bachelor of Science in Engineering, Posts and Telecommunications Institute of Technology, Hanoi, Vietnam (PTIT), 2017 – 2022.

- Major: Information Techonology
- GPA: 3.30/4.0
- Thesis (9.6/10): Joint Learning of Intent Detection and Slot Filling.
- Supervisor: Dr. Ngo Xuan Bach, Associate Professor, Vice Dean of Faculty of Information Technology, Posts and Telecommunications Institute of Technology (PTIT), Hanoi, Vietnam. (*Personal Home-page*)
- Awards and Prizes:
  - 1st prize in calculus math at MATH OLYMPIAD FOR STUDENTS, 2018 organized by Vietnam Mathematical Society, Vietnam Unions Of Science And Technology Associations.
  - 1st prize at ACM ICPC – PTIT, 2020 organized by PTIT.
  - Samsung Algorithm Certificate, sponsored and trained by Samsung & PTIT.

SKILL

- Programming Languages: Python, Java, C++.
- Software Development: Linux, Docker, Elasticsearch.
- Deep Learning Frameworks and Libraries: Pytorch, Tensorflow, scikit-learn, Huggingface, Triton Server, NVIDIA NeMo Megatron

Publications

International Refereed Conference Papers

- Nguyen Anh Tu, **Duong Xuan Hieu**, Tu Minh Phuong, Ngo Xuan Bach. A Bidirectional Joint Model for Spoken Language Understanding. In *Proceedings of the 48th IEEE International Conference on Acoustics, Speech, and Signal Processing* (ICASSP), Greece, June 2023.